Matteo Pignataro

matteo.pignataro@tiscali.it Redacted
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PROFESSIONAL SUMMARY

Software engineer with hands-on experience in C/C++, embedded systems, real-time software and robotic platforms. Experienced in low-level communication protocols, ROS 2, Docker, and hardware-software integration through autonomous racing and experimental rocketry projects. Strong background in low-level development, system integration and multidisciplinary technical coordination.

EDUCATION

Master of Computer Science and Engineering at Politecnico di Milano *09.2022 - 12.2025*

- Final Grade: 110/110 with honors
- Thesis Title: A Scalable and Bandwidth-Aware Telemetry Framework for Autonomous Racing Vehicles

Bachelor of Computer Science and Engineering at Politecnico di Milano *09.2019 - 09.2022*

- Final Grade: 105/110

High School Diploma in Computer Science at I.T.I.S. Stanislao Cannizzaro *09.2014 - 06.2019*

- Final Grade: 100/100 with honors

WORK EXPERIENCE

Researcher, System Integrator at PoliMOVE Racing Team *02.2025 - Now*

Working as a System Integrator for the PoliMOVE autonomous racing team. Developed a bandwidth-aware telemetry system with congestion detection and avoidance. Experienced frequent international travel including extended technical trips and competitions abroad. On July 24th 2025, the team achieved a historic milestone by securing the first place in the inaugural American road-course autonomous race at Laguna Seca Speedway in California.

Key technical contributions:
- Developed and integrated multithreaded software within a complex ROS2 project for autonomous vehicles;
- Designed and deployed Docker-based development environments integrated with GitLab CI/CD pipelines;
- Designed and developed a fully code-generated C++ CAN bus serialization and deserialization layer from DBC specification files, validated using automatically generated Catch2 unit tests achieving high code coverage;
- Designed a custom network bandwidth-aware protocol for telemetry;
- Performed code reviews and contributed to merge request discussions to ensure fail-safe software quality and consistency.

Project Manager at Skyward Experimental Rocketry *09.2023 - 10.2024*

Worked as Project Manager for the Lyra rocket, coordinating a multidisciplinary team of technical leaders in designing and building a paraffin-based sounding rocket. Managed project deadlines and coordinated the work of 150 people to ensure the rocket was competition-ready. On September 14th, the rocket was successfully test-launched in Roccaraso, performing as expected and validating its design. Served as team leader at the European Rocketry Challenge (EuRoC) in Portugal, where the team secured first place, winning the competition with a nominal flight.

Software Team Leader at Skyward Experimental Rocketry *09.2022 - 09.2023*

Worked as Flight Software Team Leader for the Gemini Project. Managed a team of 15 people and worked closely with the electronics department to review hardware specifications and schematics. Studied, developed, and implemented a custom communication protocol between the Ground Station and the Ground Segment Equipment to guarantee collision avoidance for low-data-rate, half-duplex radio communications.

Key achievements:
- Designed and developed the flight software for the STM32F767 microcontroller;
- Developed a CAN bus network connecting four computers, validated using hardware-in-the-loop (HIL) simulation;
- Debugged low-level issues using oscilloscopes and logic analyzers;
- Learned the dynamics of the design process as part of the technical team;
- Translated hybrid rocket engine operational requirements into software and system requirements;
- Led the launch procedures for the hybrid rocket as flight controller.

Software Team Member at Skyward Experimental Rocketry *10.2021 - 09.2022*

Worked as part of the software and control team for the Pyxis Project. Programmed the flight software for the Payload bay computer and worked with the Recovery team during parafoil testing for flight characterization and control algorithm tuning. Contributed to the team's first-place at the European Rocketry Challenge competition, achieved with a nominal flight.

Key achievements:
- Worked with an embedded system developed around the STM32F429 microcontroller;
- Developed real-time software using the Miosix RTOS and performed low-level debugging with GDB;
- Deployed flight software with standard communication protocols: SPI, UART, I2C, and CAN;
- Developed low-level device drivers from hardware datasheets and reference manuals.

SKILLS

Programming: C, C++, Python, Java, Git

Languages: Italian, English

Other: CAD, Electronics, Hand Soldering, Docker

CERTIFICATES

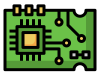
Test of English for International Communication (TOEIC): C1 (955 points) *04.2022*

Cambridge First Certificate (FCE): B1 *05.2018*

INTERESTS

Computer Graphics

Strong interest in computer graphics. Studied and used the Vulkan and OpenGL APIs to write a low-level, object-oriented framework for them. Also developed a 3D visualizer for rocket trajectories and attitude data to monitor flights from the ground station.



Electronics

Passionate about analog and digital electronics. Studied the fundamentals of electronics at university and completed a course in digital circuit design. Additionally, obtained an amateur radio operator license. Presented twice at the Rome Maker Faire showcasing two high-voltage projects. Through these experiences, learned the usage of essential electronics instrumentation such as oscilloscopes, logic analyzers, and function generators, which enabled me to design and build analog amplifiers and multiple solid-state Tesla coils at home.